

GETTING STARTED WITH AZURE OPENAI SERVICES

➔ Introduction to Azure openAI services:

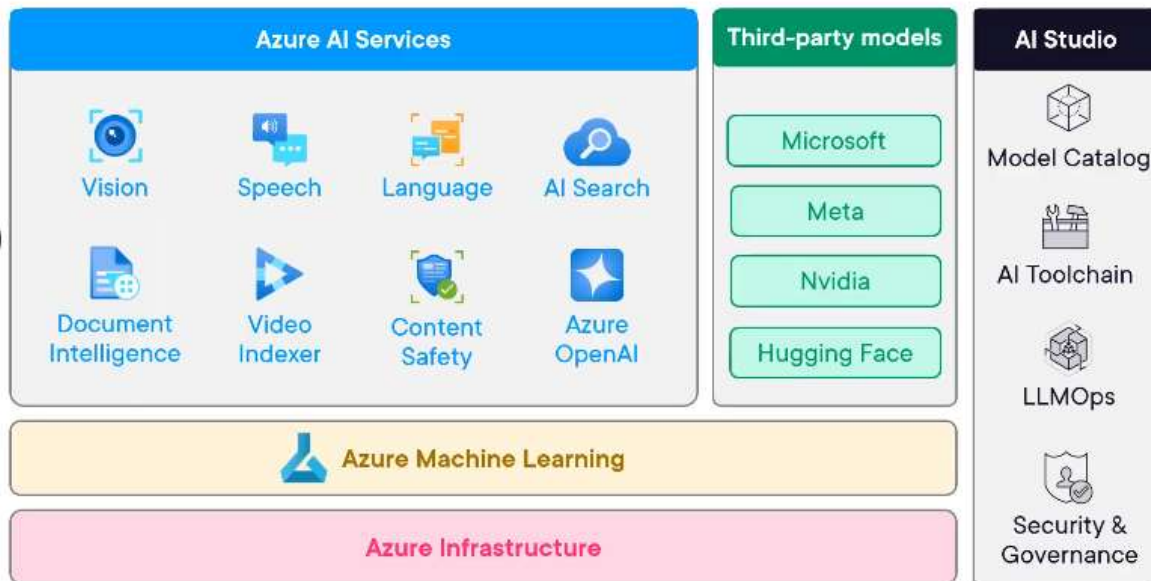
- **Generative AI** focuses on creating new content based on user input
 - It can generate Content includes text, images code, audio, videos, etc.
 - **Generative Capabilities:** These systems leverage large language models (LLMs) and other generative models to create text, images, code, and other content based on input
- Open AI :
 - Builds & provides Generative AI models
 - Supports multiple workloads
 - -Natural language, images, code, speech, embeddings, etc.
 - Models are pre-trained up to a certain date
 - Allows you to bring your own data
 - Some models can be fine-tuned

OpenAI Models

Models overview	
The OpenAI API is powered by a diverse set of models with different capabilities and price points. You can also make customizations to our models for your specific use case with fine-tuning .	
MODEL	DESCRIPTION
GPT-4o	Our high-intelligence flagship model for complex, multi-step tasks
GPT-4o mini	Our affordable and intelligent small model for fast, lightweight tasks
GPT-4 Turbo and GPT-4	The previous set of high-intelligence models
GPT-3.5 Turbo	A fast, inexpensive model for simple tasks
DALL·E	A model that can generate and edit images given a natural language prompt
TTS	A set of models that can convert text into natural sounding spoken audio
Whisper	A model that can convert audio into text
Embeddings	A set of models that can convert text into a numerical form
Moderation	A fine-tuned model that can detect whether text may be sensitive or unsafe
GPT base	A set of models without instruction following that can understand as well as generate natural language or code
Deprecated	A full list of models that have been deprecated along with the suggested replacement

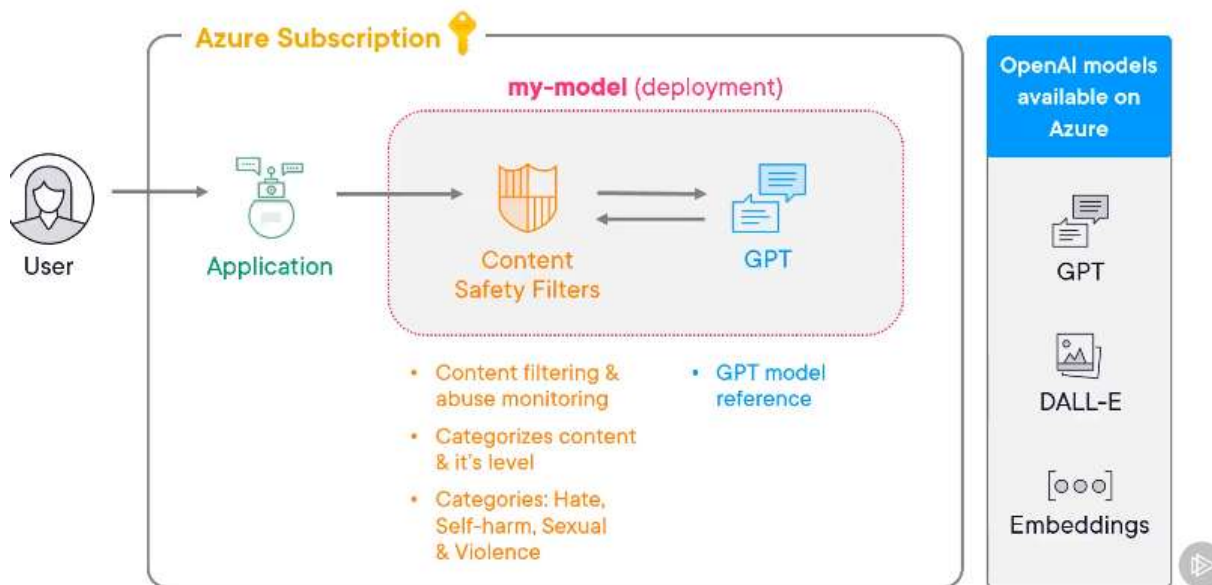
- Azure AI Portfolio:
- Azure provides various ai services and their posted on ai infrastructure

Azure AI Portfolio



- the **Azure AI Portfolio**, which consists of three main pillars for building AI applications: [1]
- **Azure AI Services**: A collection of standalone, production-ready APIs for adding specific capabilities like vision, speech, language, or search to your applications. [1]
- **Third-party Models**: Access to various models from providers such as Meta, Nvidia, Hugging Face, and Microsoft themselves. [1]
- **Azure AI Studio**: A comprehensive platform designed for end-to-end development, offering tools for orchestration, grounding models with your own data, evaluation, and security. [1, 2]
- The underlying infrastructure supporting these services is **Azure Machine Learning** and broader **Azure Infrastructure**

Azure OpenAI



- **Content Safety Filters:** A crucial component that screens both user input and the model's generated output to ensure compliance with policies. [1]
- **Abuse Monitoring:** The system automatically categorizes and monitors content for violations related to hate speech, self-harm, sexual content, and violence. [1]
- **Model Availability:** The architecture supports the deployment of various OpenAI models, including GPT for language tasks, DALL-E for image generation, and Embedding models for data analysis

Azure OpenAI

OpenAI models are available on Azure

Models are deployed in your own subscription

All your data remains within your subscription

- Enterprise data, instructions, inputs, outputs, etc.
- No data is used for retraining OpenAI models

Built-in responsible AI and content safety filters

Enterprise grade security using Microsoft Entra ID

- Provides RBAC (role-based access control) and private networks

Integration with other Azure AI services

- **Data Privacy:** Your data, including inputs and outputs, remains within your Azure subscription and is not used to train or retrain OpenAI models. [1, 2]

- **Security Controls:** The service utilizes Microsoft Entra ID for enterprise-grade security, featuring Role-Based Access Control (RBAC) and private network options. [1, 2]
- **Content Filtering:** It includes built-in responsible AI and content safety filters

How to create Azure openAI Services:

- ➔ Open Azure ai services platform
- ➔ Search azure openai and select it
- ➔ There you will be able to see all the different type of data and fill in the details
- ➔ Then fill all the details

Create Azure OpenAI

These models can be easily adapted to your specific tasks, including but not limited to content generation, summarization, image understanding, semantic search, and natural language to code translation. Top use cases include Call Centers, Virtual Assistants, Accessibility, Content Generation, and Code Development. The service also features the Assistants API, Fine Tuning capabilities and many ways to connect your data to the service for conversational experiences. The service can be scaled through Standard (tokens) and Provisioned (PTUs) deployment types.

[Learn more](#)

Project Details

Subscription * ⓘ MSDN Platforms

Resource group * ⓘ (New) AzureOpenAIRG [Create new](#)

Instance Details

Region ⓘ East US

Name * ⓘ ps-azopenai-eastus ✓

Pricing tier * ⓘ Standard S0

[View full pricing details](#)

Content review policy

[Previous](#) [Next](#)

- ➔ Wait till the instance is deployed
- ➔ Then Azure ai studio → open model catalog(100 of models)
- ➔ Select Azure openai model that take us to the space where we ever be able to see again

Deploying Openai models using Azure AI studi

- ➔ Open Azure AI studio

➔ Create a new project and name it and fill all the details

The screenshot shows the 'Create a hub for your projects' form. On the left, a sidebar contains three steps: 'Project details' (checked), 'Create a hub' (active), and 'Review and finish'. The main form area is titled 'Create a hub for your projects' and includes a description: 'A hub is the collaboration environment for your team to share your project work, model endpoints, compute, connections, and security settings. [Learn more](#)'. Below this, a question asks: 'Do you need to customize security or the [dependent resources](#) of your hub? [Go to Azure Portal](#)'. The form contains several dropdown menus: 'Hub name' (filled with 'Pluralsight-Hub'), 'Subscription' (filled with 'MSDN Platforms'), 'Resource group' (filled with 'AzureOpenAIRG'), 'Location' (filled with 'East US'), 'Connect Azure AI Services or Azure OpenAI' (filled with 'ps-azopenai-eastus'), and 'Connect Azure AI Search' (filled with 'Skip connecting'). Each dropdown has a 'Create new' link to its right. At the bottom, there are 'Back', 'Next', 'Create a project', and 'Cancel' buttons.

➔ In the project you can create new deployments

The screenshot shows the 'Select a model' dialog. The title bar says 'Select a model'. Below the title, it says: 'Choose a model to create a new deployment. For flows and other resources, create a deployment from their respective list. [Go to model catalog](#)'. There are filters for 'Models: 1787', 'Collections', 'Deployment options', and 'Inference tasks'. A 'Show description' toggle is on the right. The main area is a list of models with a search bar at the top. The models listed are: 'gpt-4o-realtime-preview' (audio-generation), 'openai-whisper-large-v3' (Speech recognition), 'openai-whisper-large' (Speech recognition), 'gpt-4' (Chat completion), 'gpt-35-turbo' (Chat completion), 'o1-preview' (Chat completion), and 'o1-mini' (Chat completion). Each model has a radio button to its right. On the right side of the dialog, there is a large area with a blue icon and the text 'Select a model to see description'. On the left, a sidebar shows various navigation options like 'Model catalog', 'Model bench', 'Prompt catalog', 'AI Services', 'Project playground', 'Chat', 'Assistants', 'Real-time audio', 'Images', 'Completions', 'Tools', 'Code', 'Prompt flow', 'Evaluation', 'Fine-tuning', 'Components', 'Data', and 'Indexes'.

→ Deploy models

Deploy model gpt-4o

Deployment details Collapse

Deployment type

Global Standard

Global (Standard): Pay per API call with higher rate limits. Traffic is routed globally and does not adhere to Azure data residency promises. Recommended starting point for most scenarios except those with data residency requirements. [Learn more about Global deployment type](#)

Model version

2024-05-13 (Default)

Connected Azure OpenAI resource

ps-azopenai-eastus

350K tokens per minute quota available for your deployment

Tokens per Minute Rate Limit

60K

Corresponding requests per minute (RPM) = 360

Content filter

DefaultV2

[Click for more](#)

Deploy **Cancel**